



Transform Yourself

ECG Signal Classification for Arrhythmia Detection Using LSTM and GRU Networks

SDG Goal3 – Good Health and Well-being

Project Guide

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Assistant Professor

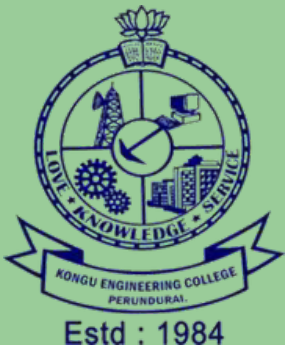
Department of AI

Project Members

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MONISH M (24ADL190)



INTRODUCTION

- ECG (Electrocardiogram) is a medical test used to measure the electrical activity of the heart.
- Early detection of arrhythmia is very important to prevent serious heart diseases.
- Artificial Intelligence (AI) can be used to automatically analyze ECG signals and detect abnormalities faster and more accurately.



PROBLEM STATEMENT

- Cardiovascular diseases are one of the leading causes of death worldwide.
- ECG signals are widely used to detect heart abnormalities.
- Traditional ECG analysis is done manually by doctors.
- Manual interpretation is time-consuming and requires expert knowledge. Arrhythmia from ECG signals
- There is a high chance of human error and delayed diagnosis.
- Hence, an automated system is required to accurately detect arrhythmia from ECG signals.



LITERATURE SURVEY

JOURNAL	YEAR	TITLE	METHODOLOGY	METRICS
IEEEXplore	2025	ECG Arrhythmia Detection Using LSTM Networks	LSTM (Long Short-Term Memory)	Accuracy: 96.1%
IEEEXplore	2024	Automated ECG Classification Using GRU	GRU (Gated Recurrent Unit)	Accuracy: 95.2%
Springer	2023	Deep Learning-Based ECG Signal Classification	CNN (Convolutional Neural Network)	Accuracy: 93.8%

Estd : 1984

LITERATURE SURVEY

JOURNAL	YEAR	TITLE	METHODOLOGY	METRICS
IEEEExplore	20223	Hybrid CNN-LSTM Model for ECG Analysis	CNN + LSTM Hybrid Model	Accuracy: 97.3%
Elsevier	2022	Machine Learning Techniques for ECG Classification	SVM + Random Forest	Accuracy: 89.6%
Wiley	2020	Real-Time ECG Monitoring and Arrhythmia Detection	Deep Learning (LSTM + GRU)	Accuracy: 97.8%

RESEARCH GAP

- Existing models like LSTM and GRU provide good accuracy but require high computational cost.
- CNN models capture features but fail to learn long-term temporal dependencies.
- Many systems are not suitable for real-time ECG monitoring.
- Traditional machine learning methods show lower accuracy compared to deep learning.
- Existing models are sensitive to noise in ECG signals.

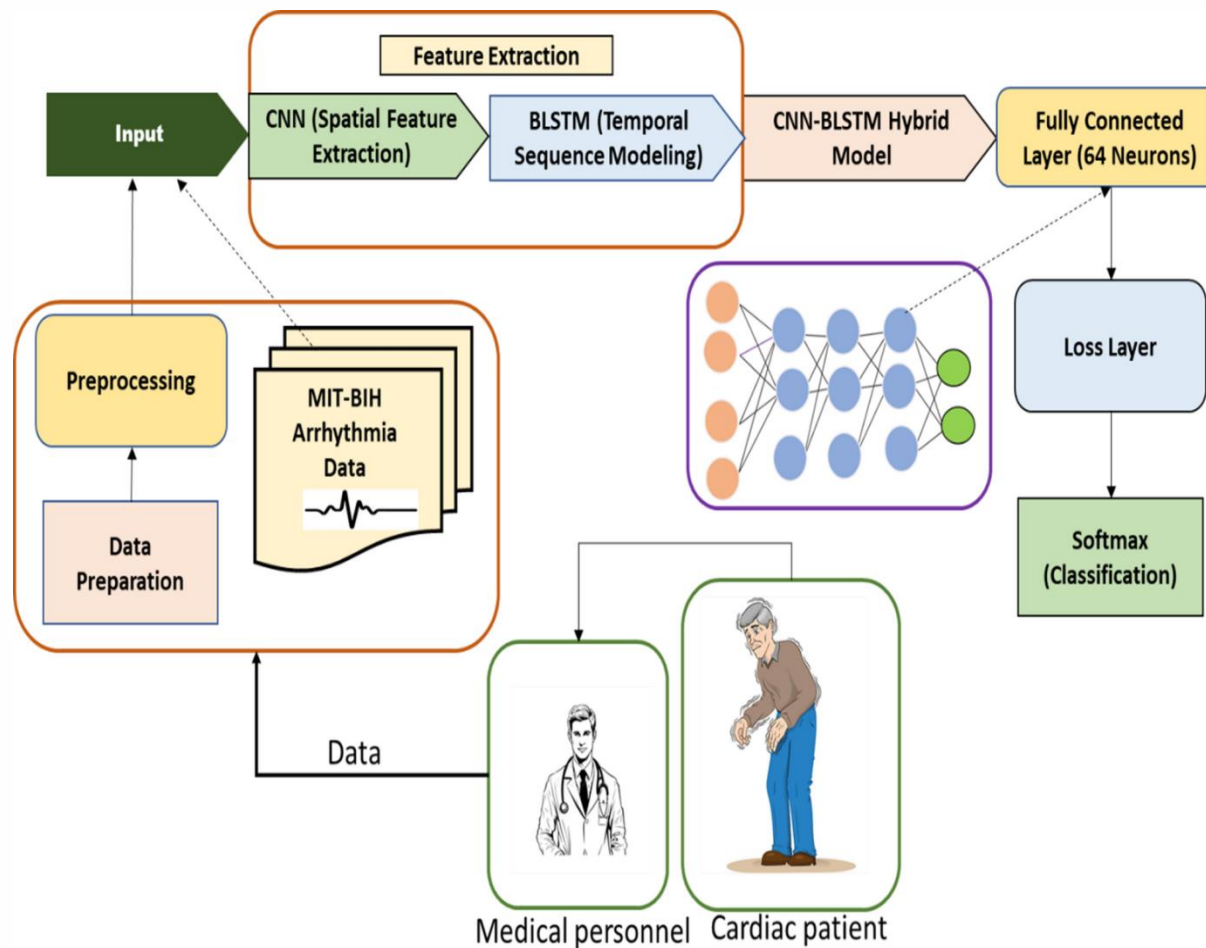
OBJECTIVE

- To implement deep learning models such as LSTM and GRU
- To analyze ECG (Electrocardiogram) signals for heart activity
- To detect abnormal heart rhythms (arrhythmia)
- To compare model performance and accuracy
- To develop an automated ECG classification system
- To classify ECG signals into multiple categories

PROPOSED WORK

- Collect ECG signal data from MIT-BIH dataset
- Perform preprocessing (segmentation, normalization, encoding)
- Convert ECG signals into time-series input format
- Apply deep learning models (LSTM and GRU)
- Train models to classify different heartbeat categories
- Evaluate model performance using accuracy and confusion matrix
- Compare results and select the best model

ARCHITECTURE DIAGRAM



DATASET DESCRIPTION

- The project uses the MIT-BIH Arrhythmia Database.
- It is a standard ECG dataset used for arrhythmia detection research.

Dataset Features

- ECG recordings from multiple patients
- Normal and abnormal heartbeat classes
- Annotated heartbeat labels
- Suitable for deep learning applications

MODULES EXPLANATION

Data Collection

Collect ECG signals from MIT-BIH dataset.

Preprocessing

Noise removal and signal normalization.

Model Training

Train CNN/LSTM model using ECG data.

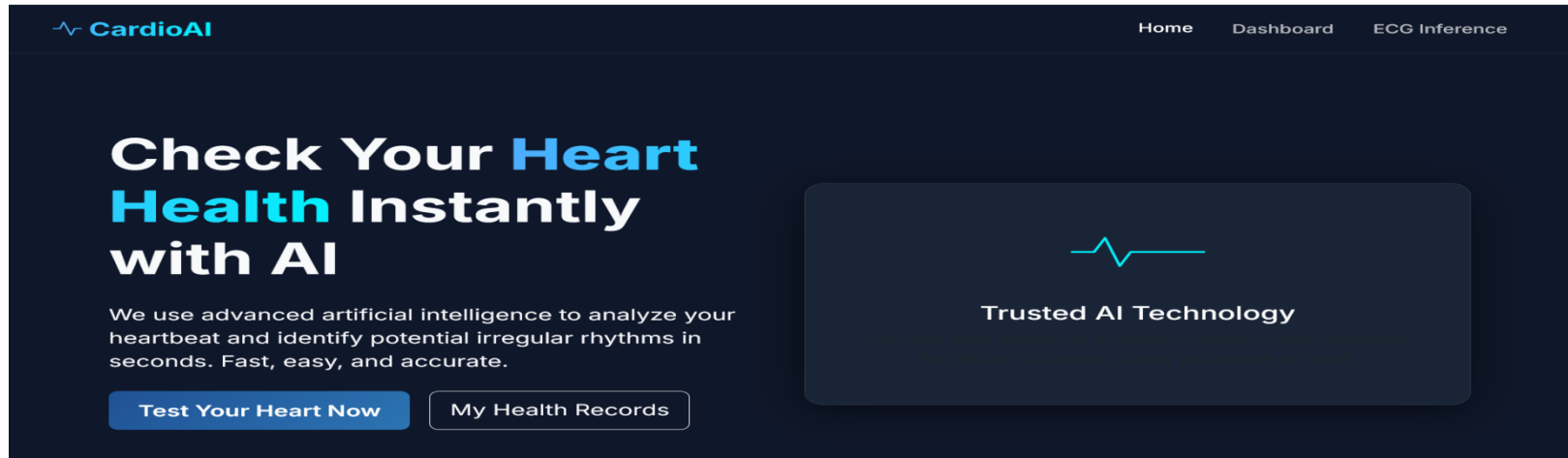
Prediction

Classify ECG signal as normal or abnormal.

Result Visualization

Display prediction results in frontend interface.

DEMO SCREENSHOT



The screenshot shows the CardioAI Home page. The header includes the CardioAI logo and navigation links for Home, Dashboard, and ECG Inference. The main content area features a large heading "Check Your Heart Health Instantly with AI" and a subheading "We use advanced artificial intelligence to analyze your heartbeat and identify potential irregular rhythms in seconds. Fast, easy, and accurate." Below this are two buttons: "Test Your Heart Now" and "My Health Records". To the right, there is a dark blue box with a teal ECG line icon and the text "Trusted AI Technology".

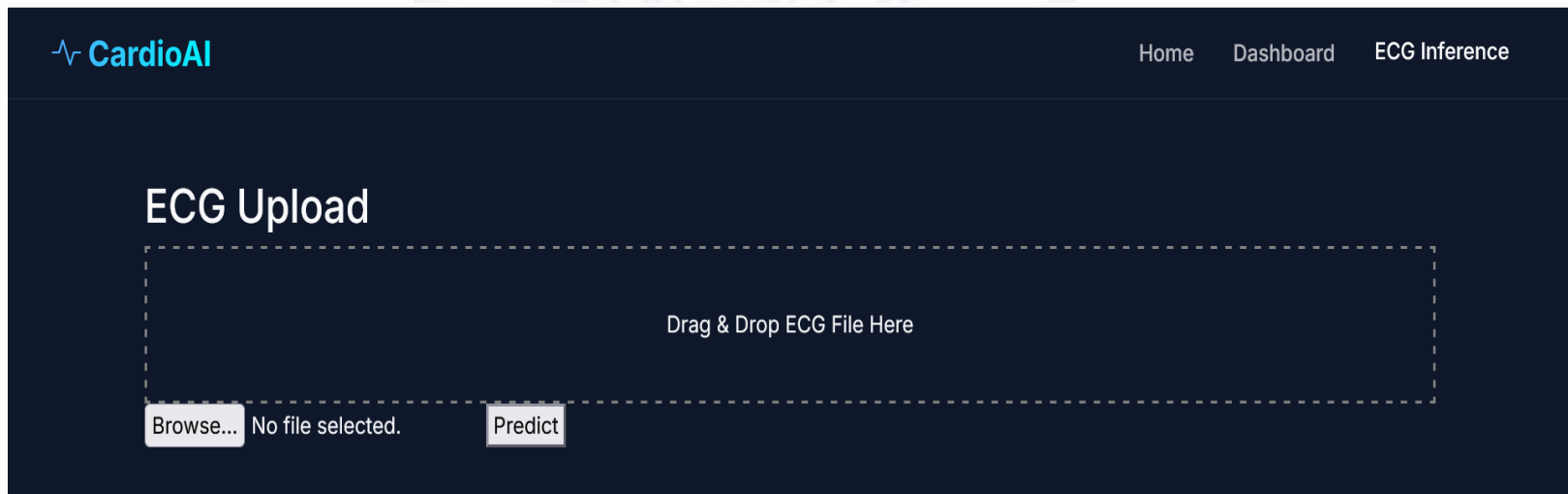
CardioAI Home Dashboard ECG Inference

Check Your Heart Health Instantly with AI

We use advanced artificial intelligence to analyze your heartbeat and identify potential irregular rhythms in seconds. Fast, easy, and accurate.

Test Your Heart Now My Health Records

Trusted AI Technology



The screenshot shows the CardioAI ECG Upload page. The header includes the CardioAI logo and navigation links for Home, Dashboard, and ECG Inference. The main content area features a heading "ECG Upload" and a large dashed box for file upload. Below the dashed box is the text "Drag & Drop ECG File Here". At the bottom, there are three buttons: "Browse...", "No file selected.", and "Predict".

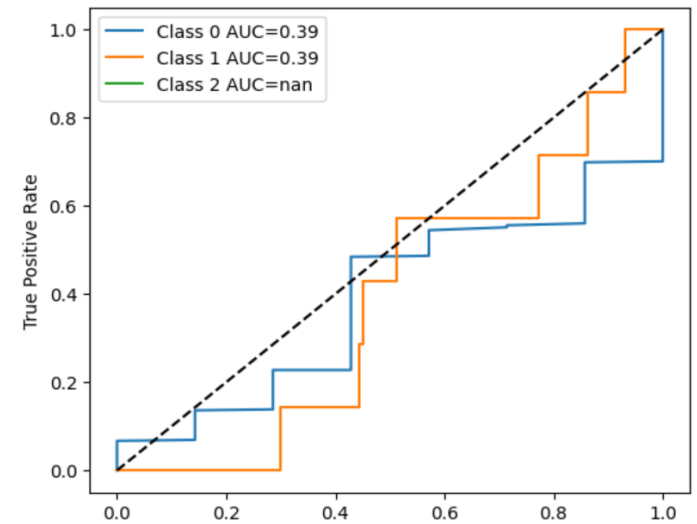
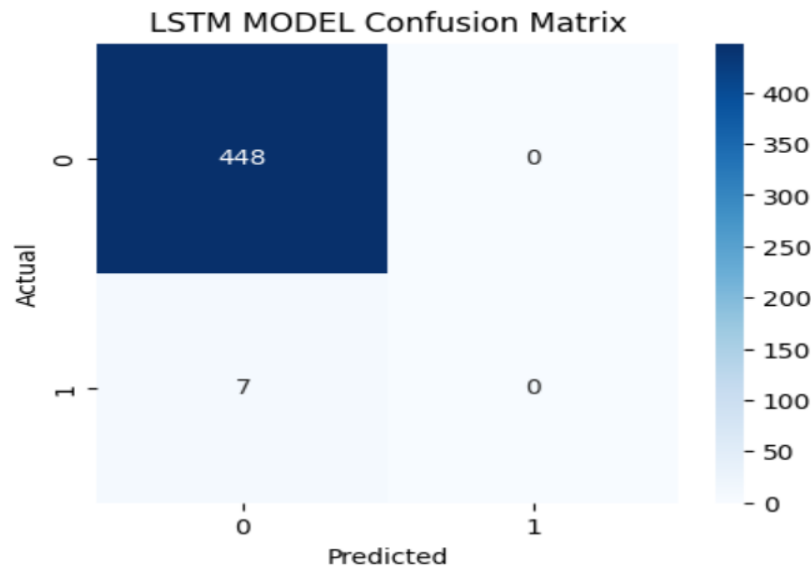
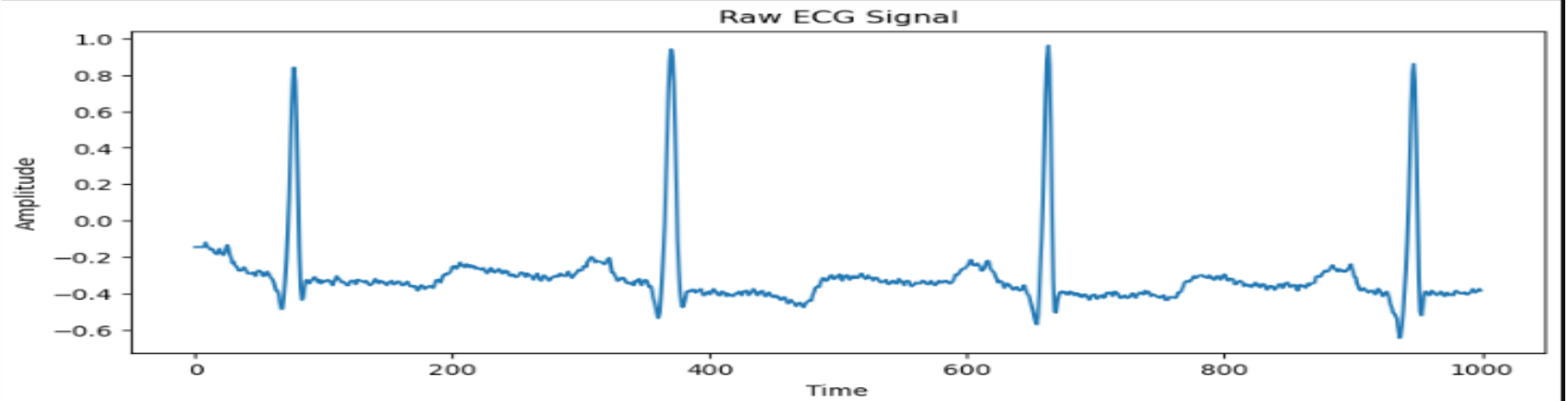
CardioAI Home Dashboard ECG Inference

ECG Upload

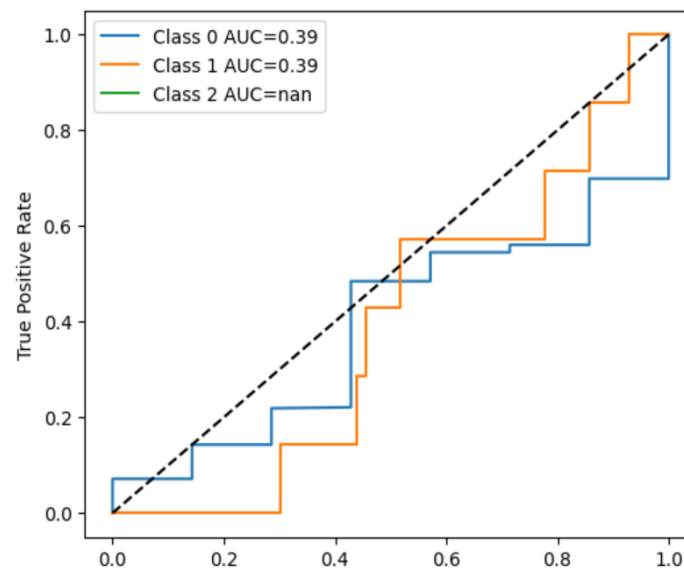
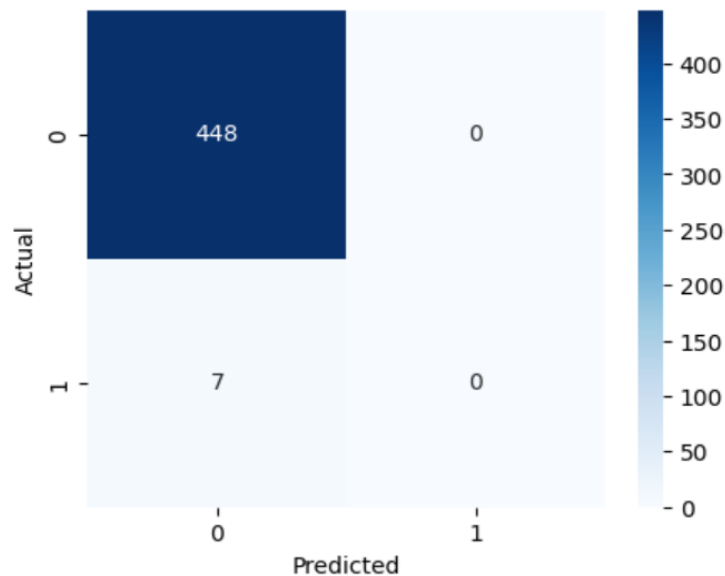
Drag & Drop ECG File Here

Browse... No file selected. Predict

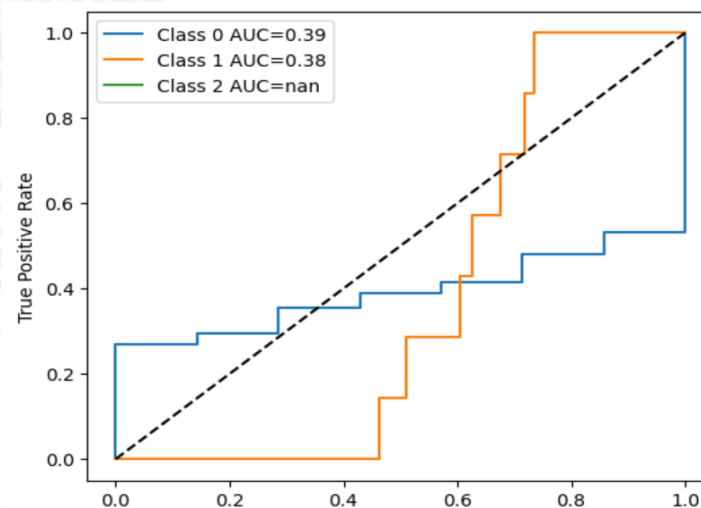
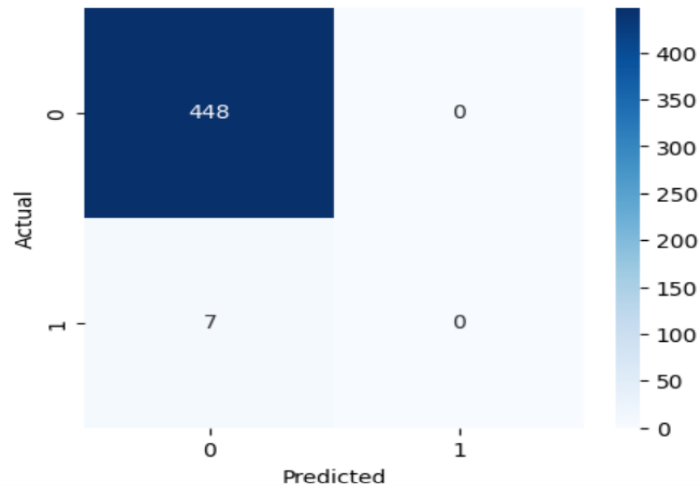
SCREENSHOT



GRU MODEL Confusion Matrix



CNN + LSTM HYBRID Confusion Matrix



PERFORMANCE METRIXS

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LSTM MODEL

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15/15	1s 60ms/step			
	precision	recall	f1-score	support
0	0.98	1.00	0.99	448
1	0.00	0.00	0.00	7
accuracy			0.98	455
macro avg	0.49	0.50	0.50	455
weighted avg	0.97	0.98	0.98	455

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GRU MODEL

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15/15	1s 67ms/step			
	precision	recall	f1-score	support
0	0.98	1.00	0.99	448
1	0.00	0.00	0.00	7
accuracy			0.98	455
macro avg	0.49	0.50	0.50	455
weighted avg	0.97	0.98	0.98	455

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CNN + LSTM HYBRID

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REFERENCES PERFORMANCE

- MIT-BIH Arrhythmia Database, “MIT-BIH Arrhythmia Database,” PhysioNet. Available: <https://physionet.org/content/mitdb/>
- Sepp Hochreiter and Jürgen Schmidhuber, “Long Short-Term Memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 2021.
- Kyunghyun Cho et al., “Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation,” 2024.

Thank You

